

Forecasting Annual U.S. Honey Bee Colony Numbers Using Time Series Data

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November 9, 2022

1 Problem Statement

North America is home to nearly 4,500 species of bees [1]. Most species are solitary, but there are 49 known species of primitive eusocial bumble bees in the United States. 41 of these species are also found in Canada, and an additional 11 species are found in Mexico. In 1622, the highly eusocial western honey bee (*Apis mellifera*) was introduced to North America from Europe and Africa [2], and it is the only species of honey bee in North America.

Honey bees and other plant pollinators are vital to the health of agricultural and plant ecosystems. In fact, honey bees pollinate many of the fruits, vegetables, and nuts the human population grows and consumes. 75% to 80% of the world's crops depend on pollinators like the honey bee [3], and the need for honey bee pollination services has increased as the demand for insect-pollinated food has risen. Unfortunately, honey bees in the United States have a high mortality rate. "...over the last decade, the U.S. annual honey bee colony mortality rate has fluctuated between 30-45%, with some beekeepers losing half or more of their colonies every year" [4]. Nonetheless, colony loss rates remain separate from honey bee population decline. The honey bee population (i.e., the overall number of colonies) has remained steady over the years, but the beekeeping and agricultural industry do face challenges [5]. There is still notable concern for the overall health of honey bee colonies, which has led to the establishment of nonprofit organizations, such as "The Bee Informed Partnership", to help improve the health and long-term sustainability of honey bees.

In the United States, honey bees have such a high mortality rate due to a complex suite of factors. "Several critical factors threaten honey bees including poor nutrition due to land use change, pesticide exposure, and introduced pests and diseases" [4]. Varroa mites, which were introduced into the United States in the 1980s, are one of the major external parasitic threats that continue to negatively impact US honey bee colonies. Elevated pathogens and pesticides also have substantial influence on honey bees, with a wide array of pesticides being detected in wax and pollen [4]. These external threats then make honey bee colonies more susceptible to serious diseases, like "Colony Collapse Disorder" (CCD), which is a phenomenon that occurs when thriving colonies empty within a few weeks because many worker bees disappear [6].

Moreover, climate change continues to have intensifying weight on honey bees, especially through more extreme temperatures. To elaborate, honey bees are active when it is 60.8 degrees Fahrenheit, but they stop flying when the temperature falls below 50 degrees. If the temperature drops below 28 degrees, then they die within one hour [7]. In a survey performed from 2014 to 2019, the mortality rate for bees was 53.5% during the winter [8]. While honey bees can survive in both high and low temperatures, they will still die if there are prolonged instances of substantially high or low temperatures because they suffer severe water loss and weakened muscles to fly [7].

Finally, air quality can damage honey bee colonies. A 2020 study performed in India demonstrated that the number of bees visiting flowers reduced significantly in areas with elevated levels of pollution, while

the opposite was true in areas where pollution was lower [9]. The study identified high and low pollution areas by investigating the levels of particulate matter (PM), consisting of sulfate, nitrates, ammonia, sodium chloride, black carbon, mineral dust, and water, in the air [9]. Note that elevated levels of PM_{2.5} means that the PM is small enough to enter the systems of humans and animals [10]. Other environmental exposures such as ozone, nitrogen, and sulfur dioxide can also have serious outcomes on honey bees, and elevated levels of pollutants can cause bees to die.

Consequently, such highly volatile dynamics in honey bee mortality can lead to unpredictable outcomes for our ecosystem and agricultural industry. Colony loss directly affects the honey bee population, and a potential decrease in the number of honey bee colonies could negatively impact crop production yields. While beekeepers can help to create more colonies, replacing colonies still comes at a great cost, and it should not be underappreciated [4]. Over the last 20 years, annual loss rates remain high, so beekeepers are consistently under pressure to offset these losses by creating new colonies [11]. Instead of reactionary decision making based on constant fluctuations in the honey bee population, it could help beekeepers, farmers, etc. to have insight into how many colonies they can expect to have in the future. This would enable the appropriate policy makers to improve assessments of local and regional habitat suitability and ultimately make more informed decisions. If these groups had advanced knowledge of substantial colony loss in the future, then perhaps they could take steps in the present to mitigate some of the problems through targeted initiatives. Therefore, the goal of this project is to utilize past data to help build a machine learning model that forecasts annual US honey bee colony numbers.

2 Literature Survey

As explained, the topic of honey bee population loss is an important one. Most of the research being done is focused on analyzing data collected through surveys. The “Bee Informed Partnership” created the “Colony Loss and Management Survey,” which monitors managed honey bee colony loss rates and identifies risk factors and protective measures [12]. Furthermore, the “National Honey Bee Survey” is a comprehensive examination of colony health throughout the United States, and it helps document honey bee diseases, pests, and pathogens [13]. The data collected through these surveys is valuable to both beekeepers and researchers.

For instance, a review published in *EcoHealth* [14] utilized data collected through surveys from individual beekeepers to study the drivers of decline in bee colonies. The study was conducted by identifying socioeconomic, political, and environmental factors affecting the bee industry, and various models were run to analyze bee colony numbers. It was found that how the bees are managed, along with pathogens and pesticides, were significant contributors to the decline of the bee colonies. The approach used in the study can be improved by adding more datasets and evaluating data across the entire United States to get a more extensive view of the factors impacting honey bee colonies, such as weather and air quality, which our project intends to do. Most importantly, our primary goal is to create a machine learning model to forecast future honey bee colony numbers.

However, there are other studies that have taken a more theoretical approach. Chen et al. [15] formulated theoretical mathematical equations to study the impact on the number of honeybee colonies, and they considered several factors as variables, such as diseases, pesticides, and food availability. Though, the equations were only based on theory, and they were not validated for practical use. The authors did state that the use of machine learning to validate the theoretical models was necessary since their approach lacked the use of real data. Our project will use machine learning modeling to validate real historical time series data, and we’ll consider various factors when building our machine learning models.

Regardless, there are still only a few studies related to forecasting honey bee activity or colony numbers.

Ngo et al. [16] used temporal convolutional neural networks to forecast the population daily loss rate of honey bee colonies, and they determined warning levels based on the predictions. Yet, their project was narrowly focused on individual honey bee colonies since they collected experimental data from a honey bee farm located in Taiwan to help construct their algorithm. Moreover, Rumkee et al. [17] simulated the effects of pesticides on different honey bee life stages using the BEEHAVE model. “The BEEHAVE model simulates the development of a honey bee colony and its foraging of nectar and pollen in different landscapes” [18]. Matthias et al. [19] was another study that utilized the BEEHAVE system. The authors found the increase in failure of bee colonies were attributed to pathogens, forage availability, and potentially pesticides. To sum up, all these initiatives focused on how individual honey bee colonies responded to external stressors, specifically pesticide exposure and pathogens.

Our study, on the other hand, will have a broader focus than the research projects outlined in the previous paragraph since we will try to predict the annual honey bee population for individual states and the United States as a whole. In addition, we will try to consider several factors that are correlated with increased honey bee mortality, including pesticides, diseases, pests, temperature, precipitation, and air quality. The aim of this project is to account for the factors that have a potential impact on honey bees to make informed predictions. Finally, we will experiment with different machine learning models to see which approach produces the best forecasts of future honey bee colony numbers. predict the future bee hive colony numbers.

3 Data Sources

For our project, we produced two separate data sources to experiment with different kinds of methodologies because we want to examine which ones produce the best predictive results for future honey bee colony numbers.

3.1 Dataset A

To easily distinguish between the two datasets we are considering, we will often refer to the dataset described in this section as “Dataset A” going forward. Dataset A is a hierarchical time series that has continuous annual data on honey bee populations, which we are treating as our response variable, for 39 states and “other states” (an aggregation of all other states) from 1987 to 2021. The data was retrieved from the National Agricultural Statistics Service (NASS), which is part of the United States Department of Agriculture (USDA) [20].

Important to note, this was the most complete and expansive dataset on annual honey bee populations that could be obtained from the NASS. The NASS creates its datasets through a wide-ranging Census conducted every 5 years, or through more constrained quarterly or annual surveys. Our group decided to focus on the data produced from the annual surveys. Unfortunately, due to the limitations of surveys, there were some states, such as Connecticut, that did not have observations for the entire time horizon. So, any states that did not have continuous observations from 1987 to 2021 were grouped together in the already existent series “other states.” This helped to create 40 unique time series that all had continuous data for the entire time horizon.

With this setup, Dataset A can be categorized as a hierarchal time series. The total honey bee population in the United States represents the “top” of the hierarchy, or the most aggregate level of data. So, the t -th observation for the United States series is denoted by y_t for $t = 1987, \dots, 2021$. This top-level series can then be disaggregated into 40 separate series, one for each of the 39 states and “other states”, where $y_{j,t}$ denotes the t -th observation corresponding to the j -th series for $j = 1, \dots, 40$. For any time t , there observations for

all the j series will sum to the observation for the United states as a whole, i.e.,

$$y_t = y_{1,t} + \dots + y_{40,t}$$

Nonetheless, we were able to generate annual data pertaining to climate and air quality that can be combined with Dataset A to identify the correlations between these features and the honey bee populations. This could potentially improve the predictions of our models that utilize Dataset A. First, the climate data was retrieved from the National Centers for Environmental Information (NCEI) [21]. The dataset consists of monthly weather measurements at the state level from 1895 to the present. For our project, the maximum temperature, minimum temperature, and precipitation data were retained. During the preprocessing, the monthly data was rolled up to the annual level for each state by taking the monthly maximum and minimum temperatures, respectively, over the 12-month readings, while the annual precipitation was calculated by adding the 12-month readings. Next, the air quality data was obtained from the United States Environmental Protection Agency (EPA) [22]. The annual summary data per county was extracted for the years of 1987 to 2022, and the data was preprocessed to merge all the various data files sets into one dataset. For each feature column on air quality, the data was grouped into states and counties, and the aggregate mean for 50 states was calculated using the appropriate county values.

3.2 Dataset B

Going forward, we will often refer to the dataset described in this section as “Dataset B.” This dataset contains annual honey bee data for 45 states and “other states” (an aggregation of data from Alaska, Delaware, Nevada, New Hampshire, and Rhode Island) from 2015 to 2021. Much of the dataset was obtained from NASS quarterly survey data [20]. It not only includes data on honey bee populations, but it also has data on factors that impact honey bee mortality, such as colony renovation, diseases, pests, pesticides, and “other causes”. The quarterly data for each state was aggregated to the annual level by taking the average of the four quarters for each factor. We also obtained annual data on the percentage of beekeepers and honey bee colonies exclusive to each state, along with a weighted average of total colony loss, from the “Bee Informed Partnership” [23]. All these additional features have been combined with the annual data on air quality and climate to produce an even larger set of independent variables. Finally, the original survey data used to create Dataset A did contain annual data on honey bee populations from 2015 to 2021 for the 45 states represented in Dataset B. Thus, we simply reused this data to represent our response variable in Dataset B.

Now, the structure of Dataset B is more in line with longitudinal or panel data. It not only has data for our response variable, but it also contains data for various independent variables. Longitudinal or panel data contains a collection of various quantities obtained across multiple observational units, in this case individual states, that are assembled over even intervals in time and ordered chronologically. So, for each observational unit i in Dataset B, we have a time series $\{(x_t, y_t) : t = 1, \dots, 7\}$, x_t is a $n \times 1$ feature vector, y_t is the response variable, and we have a total of $T = 7$ years (i.e., 2015 to 2021). Thus, we can write the data for all our $m = 46$ observational units compactly in matrix form. We have $\{(\mathbf{X}^i, \mathbf{y}^i) : i = 1, \dots, 46\}$ where $\mathbf{X}^i$ is a $T \times n$, written as

$$\mathbf{X}^i = \begin{bmatrix} (x_1^i)^T \\ (x_2^i)^T \\ \vdots \\ (x_T^i)^T \end{bmatrix}$$

where x_t^i is our standard $n \times 1$ feature vector, and $\mathbf{y}^i$ a $T \times 1$ vector of response variables.

4 Methodologies

Currently, our group is still researching various machine learning algorithms to potentially experiment and evaluate. Yet, we have identified a few promising modeling approaches that we do intend to formulate and test.

4.1 Vector Autoregression (VAR)

One approach for forecasting multivariate time series is the Vector Autoregression model which can be used when two or more time series influence each other. In our project, there are multiple time series datasets that are going to be used to predict the annual bee colony numbers, so a VAR model can be used as one of the methods in predicting the bee colony numbers.

To use the VAR model, we will test for causation using Granger's Causality Test to ascertain the fact the system of series influences each other. Given that we have multiple time series we can test for cointegration which will reveal if they have long run statistically significant relationship. We will employ the Johanssen test for cointegration before running the VAR model. The VAR model also requires the time series to be stationary so the Augmented Dickey-Fuller Test will be used to test stationarity. The data will then be split into Training and Test set. After evaluating the model performance, the bee colony numbers for future years will be predicted using the VAR model.

4.2 Pooled Ordinary Least Squares (OLS) Regression Model

For Dataset B, a pooled OLS regression model can be tested. It can be described as a simple OLS model that is performed on panel data. It ignores time and individual characteristics, and it focuses only on the dependencies between the individual units. We assume a fixed number of time periods T , but the number of observational units m can grow larger. So, we want many observational units (i.e., states), and a few time periods per unit, which Dataset B satisfies.

A linear panel data model for each observational unit i in the population is

$$y_t = x_t \beta + \varepsilon_t$$

where $t = 1, \dots, T$. With the panel data setup described in the Section 3.2, we can write the model for an observational unit i in matrix form

$$\mathbf{y}^i = \mathbf{X}^i \beta + \varepsilon^i$$

where β is the $n \times 1$ population parameter that we want to estimate, and ε^i is the $T \times 1$ vector of error terms. Note that we have the same β for all t , which is not often restrictive because x_t^i can be very general. The pooled OLS estimator looks just like the single-equations OLS estimator, but $\mathbf{X}^i$ is a matrix and $\mathbf{y}^i$ is a vector:

$$\begin{aligned} \hat{\beta} &= \left(\frac{1}{m} \sum_{i=1}^m (\mathbf{X}^i)^T \mathbf{X}^i \right)^{-1} \left(\frac{1}{m} \sum_{i=1}^m (\mathbf{X}^i)^T \mathbf{y}^i \right) \\ &= \left(\sum_{i=1}^m \sum_{t=1}^T (x_t^i)^T (x_t^i) \right)^{-1} \left(\sum_{i=1}^m \sum_{t=1}^T (x_t^i)^T y_t^i \right) \end{aligned}$$

A pooled OLS regression model is often a good starting point and reference model for other models related to panel data, such as the "Fixed Effects Regression Model" or the "Random Effects Regression Model" [24].

4.3 Long Short-Term Memory (LSTM)

A time series modeling is considered a sequence-to-sequence prediction model to forecast the future response based on historical data. With the use of real data collected for this project consisting of a time series with multiple features, modeling can be used to identify correlations with various features that impact the outcome of the number of bee colonies for each state in the US. One type of modeling considered for this project is the long short-term memory (LSTM) recurrent neural network. The model can be trained with historical data, validated, and tested to forecast the number of bee colonies in the future using various dependent features. In addition, it offers the capability to optimize internally using the gradient descent for each of its parameters and equations using the loss function of mean squared error to obtain the best fit model and obtain better results. Trends, and seasonality will impact the results for a time series, and they can be detected prior to the model and ensure a stationary time series.

LSTM depends on the input sequence (order of observations) that is important for prediction. It processes the information (data) sequentially passing on information as it propagates forward in the neural network. The LSTM network can learn a mapping function for the inputs over time to an output. The internal memory can condition the output on the recent context in the input sequence along with the previous state [25]. It offers the capability to learn long term correlations in a sequence and can be modeled by multiple parallel input series separately with varied length of input sequence to produce a variable length of output sequence [25]. LSTM can remember information for longer periods of time (resolving the issue of short-term memory with recurrent neural networks) and it resolves the back propagation problem of the gradient vanishing when assigning weights to the neural networks. LSTM uses gates and cell states which are capable of learning long-term dependencies to regulate the flow of information. The cells make decisions on how much, when to store and release information to make accurate predictions.

5 Evaluations and Final Results

To assess the performances of our models, it is important to evaluate a model's forecast accuracy using genuine forecasts. We are unable to randomly assign data into training and test sets because it makes no sense to use future data to forecast values in the past. With time series data, we must consider the temporal dependencies between the observations in our two datasets. So, particular care must be taken in splitting the data to prevent data leakage. This means our data must be organized chronologically, and then we pick a point in time to split our data. All data prior to and including this point in time will be used to fit our models, while data occurring after will be withheld for testing.

Furthermore, when comparing different forecasting models, we want to pick the model with the smallest forecast error, which is the difference between an observed value and its forecast. According to Hyndman and Athanasopoulos [26], forecast error can be written as

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

where the training data is given by $\{y_1, \dots, y_T\}$ and the test data is given by $\{y_{T+1}, y_{T+2}, \dots\}$. Also, there are several different measures that can be used to evaluate forecast accuracy. Since the forecast errors will be on the same scale of data (i.e., the total number of honey bee colonies), then it is permissible to use accuracy measures that are scale dependent. Two popular versions include root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\text{mean}(e_t^2)}$$

and mean absolute error (MAE)

$$\text{MAE} = \text{mean}(|e_t|)$$

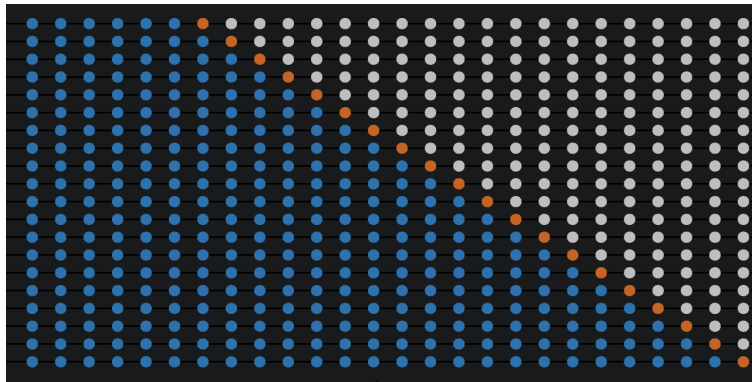
“A forecast method that minimizes the MAE will lead to forecasts of the media, while minimizing RMSE will lead to forecasts of the mean” [26]. We also need to an accuracy measure to compare the forecast performances between datasets A and B. So, mean absolute percentage error (MAPE) is another accuracy measure we will consider:

$$\text{MAPE} = \text{mean}(|100e_t/y_t|)$$

This is because “Percentage errors have the advantage of being unit-free, and so are frequently used to compare forecast performances between datasets” [26].

Moreover, a more sophisticated method than simply splitting our data into training and test sets is known as time series cross-validation (TSCV). This involves performing cross-validation on a rolling basis. For each iteration, we start with a subset of training data, forecast for a small subset of future data, and then check the accuracy of the forecasted data points. In the next iteration, we expand the training dataset by adding the data points we forecasted on from the previous iteration, and then another small subset of future data points is forecasted and evaluated. This process continues for the entire dataset. However, a fair portion of the earliest observations will always be used for training since it is not possible to obtain reliable forecasts on small training sets.

Continuing, we will primarily utilize the TSCV procedure outlined by Hyndman and Athanasopoulos [26] involving one-step forecasts. The series of test sets will consist of all the observations from a single year. The corresponding training sets will consist of all the observations that occurred prior to the observations that form the given test set. This means that no future observations will be used in constructing the forecasts, which is exactly what we want to implement. The diagram below illustrates the series of training and test sets, where the blue observations form the training sets, and the orange observations form the test sets.



Finally, the forecast accuracy is computed by averaging over the test sets. To choose the best forecasting model, we will find the model with the smallest forecast accuracy measures, such as RMSE, which will be computed using TSCV.

6 Potential Challenges

- Should we build models using both datasets A and B, or should we only focus on utilizing one of the datasets?
- If we do decide to utilize both datasets, then, in practice, is it even permitted, to compare models that are built using different datasets? If so, then what is the best way to compare the models that are built using different datasets?

- If we are truly going to compare models that are built using datasets A and B, respectively, then we need to pick a common starting point that will be our first “test set” when performing time series cross-validation
- Datasets A and B also have different sets of states represented. Dataset B has six additional states compared to Dataset A (i.e., Connecticut, Maryland, Massachusetts, Oklahoma, New Mexico, and South Carolina). So, do we eliminate these states in Dataset B, or is it possible to split out the “other states” observation in Dataset A to build out observations for these six states?
- With Dataset A, are we going to build time series models that exclusively focus on our response variable, or are we going to build models that also utilize explanatory features, such as air quality, temperature, and precipitation?

References

1. Buchman, Stephen. (2002). *The Bees of the World*. By Charles D Michener. Baltimore (Maryland): Johns Hopkins University Press. *The Quarterly Review of Biology*. 77. 334-335. [10.1086/345210](https://doi.org/10.1086/345210)
2. Sheppard, W.S. “A History of the introduction of honey bee races into the USA. II” *American Bee Journal*, 129: 664-667. Accessed on: September 2022, <https://agris.fao.org/agris-search/search.do?recordID=US9009419>
3. Nicholas W. Calderone, “Insect Pollinated Crops, Insect Pollinators and US Agriculture: Trend Analysis of Aggregate Data for the Period 1992–2009.” *PLOS ONE* 7(5): e37235, Published: May 22, 2012, <https://doi.org/10.1371/journal.pone.0037235>
4. Bee Informed Team, “Frequently Asked Questions”, Accessed Nov. 6, 2022, <https://beeinformed.org/about/frequently-asked-questions/>
5. Steven Savage, “Honeybees Are Not ‘In Decline’, But the Beekeeping Industry Does Face Challenges”, *Forbes*, Published: April 30, 2022, <https://www.forbes.com/sites/stevensavage/2022/04/30/honeybees-are-not-in-decline-but-the-beekeeping-industry-does-face-challenges/?sh=638b5076657c>
6. U.S. Environmental Protection Agency, “Colony Collapse Disorder”, *EPA*, Last modified: October 6, 2022, <https://www.epa.gov/pollinator-protection/colony-collapse-disorder>
7. Karen Allison, “Bee Temperature Tolerance: What Temperature Kills Bees?”, *Busy Beekeeping*, Published: June 22, 2022, <https://busybeekeeping.com/bee-temperature-tolerance-can-bees-survive-when-it-gets-very-hot-or-cold/>
8. Martina Calovi, Christina M. Grozinger, Douglas A. Miller, & Sarah C. Goslee, “Summer weather conditions influence winter survival of honey bees (*Apis mellifera*) in the northeastern United States.” *Scientific Reports* 11: 1553, Published: January 15, 2021, <https://doi.org/10.1038/s41598-021-81051-8>
9. Janet Pelley (special to C&EN), “Air pollution hurts honeybees”, *C&EN*, Published: August 11, 2020, <https://cen.acs.org/environment/pollution/Air-pollution-hurts-honeybees/98/web/2020/08>
10. World Health Organization, “Ambient (outdoor) air pollution”, *WHO*, Published: September 22, 2021, [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)
11. U.S. Department of Agriculture, Animal and Plant and Inspection Service, “Honey Bees”, Last Modified: Jul 25, 2022, <https://www.aphis.usda.gov/aphis/ourfocus/planthealth/plant-pest-and-disease-programs/honey-bees/honeybees>
12. Bee Informed Team, “2021-2022: Preliminary Results From The Bee Informed Partnership”, *Bee Informed Partnership Inc (BIP)*, Published: July 27, 2022,

<https://beeinformed.org/2022/07/27/united-states-honey-bee-colony-losses-2021-2022-preliminary-results-from-the-bee-informed-partnership/>

13. Honey Bee Lab at University of Maryland, “APHIS National Honey Bee Survey”, Accessed on: Nov. 6, 2022, <https://ushoneybeehealthsurvey.info/>
14. Kristine M. Smith, Elizabeth H. Loh, Melinda K. Rostal, Carlos M. Zambrana-Torrel, Luciana Mendiola & Peter Dasak. “Pathogens, Pests, and Economics: Drivers of Honey Bee Colony Declines and Losses.” *EcoHealth* 10: 434–445, Published: February 5, 2014, <https://doi.org/10.1007/s10393-013-0870-2>
15. Jun Chen, Gloria DeGrandi-Hoffman, Vardayani Ratti, Yun Kang. “Review on mathematical modeling of honeybee population dynamics.” *Mathematical Biosciences and Engineering* 18(6): 9606-9650, Published: November 4, 2021, <https://doi.org/10.3934/mbe.2021471>
16. Ngo T-N, Rustia DJA, Yang E-C, Lin T-T. “Honey Bee Colony Population Daily Loss Rate Forecasting and an Early Warning Method Using Temporal Convolutional Networks.” *Sensors* 21(11): 3900, Published: June 4, 2021, <https://doi.org/10.3390/s21113900>
17. Jack C. O. Rumke, Matthias A. Becher, Pernille Thorbek, Peter J. Kennedy, & Juliet L. Osborne. “Predicting Honeybee Colony Failure: Using the BEEHAVE Model to Simulate Colony Responses to Pesticides”, *Environmental Science & Technology* 49(21): 12879-12887, Published: October 7, 2015, <https://doi.org/10.1021/acs.est.5b03593>
18. Matthias A. Becher. “BEEHAVE simulates the development of a honeybee colony and its foraging of nectar and pollen in different landscapes”, *BEEHAVE*, Published: 2013, <https://beehave-model.net>
19. Matthias A. Becher, Grimm, Volker, Pernille Thorbek, Juliane Horn, Peter J. Kennedy, and Juliet L. Osborne. “BEEHAVE: a systems model of honeybee colony dynamics and foraging to explore multifactorial causes of colony failure”. *Applied Ecology* 51: 470-482, Published: March 4, 2014, <https://doi.org/10.1111/1365-2664.12222>
20. United States Department of Agriculture, National Agricultural Statistics Service, <https://www.nass.usda.gov/>
21. National Centers for Environmental Information, Weather data set, <ftp://ftp.ncdc.noaa.gov/pub/data/cirs/climdiv/>
22. United States Environmental Protection Agency, Air Quality Data Set, https://aqs.epa.gov/aqsweb/airdata/download_files.html
23. Bee Informed Partnership, Open Research: <https://bip2.beeinformed.org/>
24. Wooldridge, Jeffrey M. (2021) *Econometric Analysis of Cross Section and Panel Data*. The MIT Press. <https://ipcig.org/evaluation/apoio/Wooldridge%20-%20Cross-section%20and%20Panel%20Data.pdf>
25. Brownlee, Jason. (2017) *Long Short-Term Memory Networks with Python*, p 229.

26. Hyndman, R.J., & Athanasopoulos, G. (2021) *Forecasting: Principles and Practice*, 3rd edition, OTexts: Melbourne Australia. [OTexts.com/fpp3](https://otexts.com/fpp3). Accessed on: November 8, 2022